Hybrid Flexible-Rigid Neural Probe for Chronic Recording of Neural Signals with Reduced Tissue Damage

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하이브리드 유연-강성 구조를 이용하여 조직 손상을 최소화한 장기간 신경신호 측정용 뉴럴 프로브

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Abstract

This paper reports a hybrid flexible-rigid neural probe designed for chronic intracortical implantation. The proposed probe architecture is based on a serpentine flexible bridge that couples an ultra-thin silicon backbone with a soft polyimide interface, enabling the reduction of tissue damage both during implantation and after insertion. This bridged design allows multi-directional mechanical compliance along the axial, lateral, and out-of-plane directions, thereby reducing mechanical constraint between the probe and brain tissue during micromotion.

We performed in vitro and in vivo experiments to demonstrate the mechanical adaptability of the fabricated probe. The proposed hybrid design is expected to have a significant impact on chronic neural implantation by reducing tissue stress during brain movement.

Keywords: flexible-rigid neural probe, serpentine bridge, multi-directional displacement, micromotion compensation

1. Introduction

Neural probes are widely used for in vitro and in vivo investigation of neural activity in the central nervous system. While rigid penetrating probes provide reliable insertion and precise targeting, their mechanical mismatch with soft brain tissue limits long-term stability during chronic implantation [1].

Polymer-based flexible neural probes reduce the effective stiffness of the implanted substrate but often require additional biodegradable stiffening coatings for implantation, which may introduce insertion-induced damage due to an insertion shuttle or a large structure that is required for ensuring enough insertion force. Mixed silicon-polymer probes address this limitation by employing a silicon shuttle for the insertion; however, flexible ribbon designs typically provide mechanical compliance along a single direction, limiting their ability to accommodate the complex three-dimensional micromotions of brain tissue [3].

In this study, we report a new hybrid flexible-rigid neural probe featuring a serpentine flexible bridge that enables multi-directional mechanical compliance along all three spatial axes.

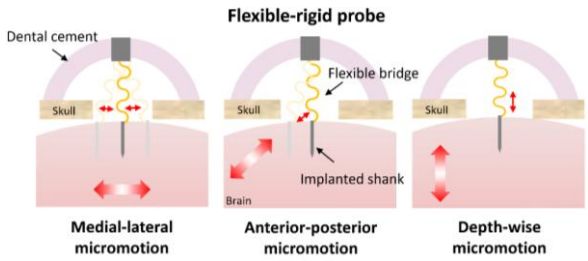

Fig. 1 A conceptual illustration of the probe implanted into the brain

The proposed architecture intentionally separates the signal interconnects into mechanically independent paths, enabling each line to move more independently in response to brain motion while maintaining a free-floating probe configuration after implantation.

2. Manuscript body

2.1 Hybrid probe architecture

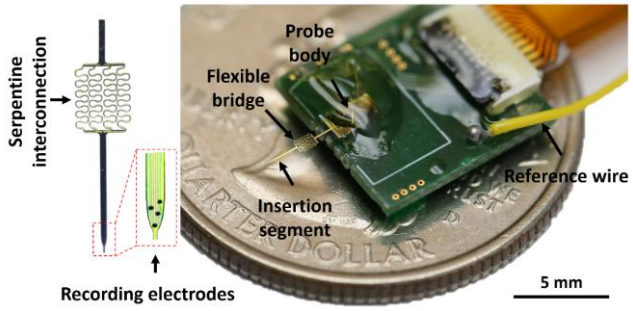

Fig. 2 Illustration of the hybrid-flexible neural probe

The hybrid flexible-rigid neural probe consisting of a rigid insertion shank, a flexible serpentine bridge, and a rigid probe body for electrical interconnection is illustrated in figure 2.

The insertion shank is formed by a 10 μm-thick silicon substrate laminated with a 7 μm-thick polyimide layer. This structure provides sufficient stiffness for insertion while maintaining a reduced cross-sectional profile. The probe is designed for chronic intracortical implantation across different brain regions. Accordingly, the insertion shank length is varied between 1 and 5 mm depending on the targeted brain area. The length of the serpentine flexible bridge is also adjusted to accommodate different

implantation geometries and skull-brain spacing across animal models, including mice, rats, rabbits, non-human primates, and humans.

Each flexible interconnect is 12 µm wide and connects a single platinum recording electrode on the insertion shank to the probe body. The serpentine geometry provides low bending stiffness and mechanical compliance along all directions.

2.2 Microfabrication process

The hybrid neural probe was fabricated using a MEMS-compatible process on a silicon-on-insulator (SOI) wafer with a 10 µm-thick top silicon layer. Following wafer cleaning, a thermal SiO_2 layer was grown to serve as an electrical insulation layer and a hard etching mask (Fig.3).

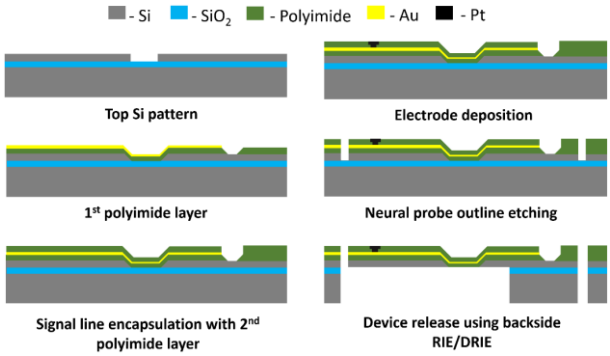

Fig. 3 Fabrication process of the proposed flexible neural probe

The top silicon layer was patterned using TMAH wet etching to define the silicon regions underlying the flexible bridge. A thin polyimide layer was then spin-coated over the wafer and fully cured to form the flexible layer. Openings for wire-bonding pads were defined by reactive ion etching (RIE), followed by metal deposition and lift-off to form the signal interconnects.

A second polyimide layer was deposited to encapsulate the metal interconnects. The serpentine bridge geometry, electrode sites, and probe shank outlines were subsequently defined by RIE and deep reactive ion etching (DRIE) of the silicon layer. Finally, the probes were released from the wafer by backside DRIE.

2.3 In vitro mechanical evaluation using agarose gel

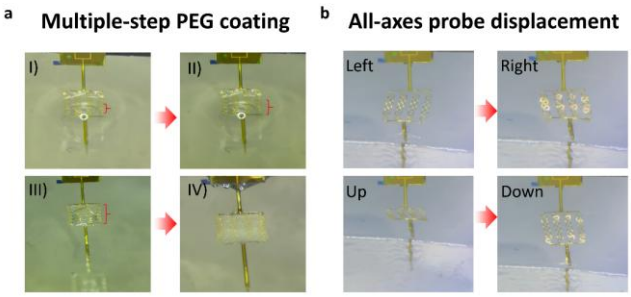

Fig. 4 PEG-assisted stiffening (a) and motion test of the implanted neural probe in agarose gel (b)

In vitro implantation tests were performed using 0.6% agarose gel as a brain-mimicking phantom. To facilitate insertion, only the serpentine flexible bridge was temporarily stiffened by multiple dip-coating cycles in polyethylene glycol (PEG), while the rigid insertion shank remained uncoated.

After insertion to the target depth, the PEG coating was dissolved by applying water droplets (Fig. 4a), exposing the flexible bridge on top of the gel. The agarose block was then mechanically perturbed to induce multi-directional displacement, and the implanted probe followed the applied motion without structural failure, demonstrating mechanical compliance under simulated three-dimensional brain movement (Fig. 4b).

2.4 In vivo implantation and neural recording

Several animal experiments were conducted to evaluate the performance of the proposed hybrid flexible-rigid neural probe under chronic implantation conditions. The primary goal of these studies was to assess the ability of the probe to accommodate brain movement while maintaining stable neural interfacing over time.

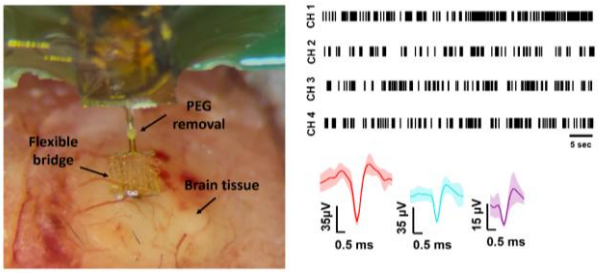

Fig. 5 Schematic illustration of the flexible neural probe implantation in the brain

Electrical signals, including neural spike activity, were collected on a weekly basis to evaluate long-term recording behavior during natural brain motion.

2.4 Conclusion

In this study, a hybrid flexible–rigid neural probe featuring a serpentine flexible bridge for chronic intracortical implantation is proposed and demonstrated. The presented architecture combines a rigid silicon insertion shank with a mechanically compliant polyimide bridge, enabling a free-floating neural interface after implantation. The results indicate that the proposed probe accommodates multi-directional brain movement while maintaining stable neural interfacing over time.

Moreover, the proposed probe is expected to be suitable for chronic neural recording across different animal models, as its mechanically adaptive architecture accommodates species-dependent brain motion, including larger-amplitude pulsations observed in larger brains.

Acknowledgments

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